Comparative Analysis of Public Transport Networks in Lisbon, Paris and Luxembourg

Andressa Litwinski
Faculdade de Ciências, Universidade do Porto
Porto, Portugal
up202103489@edu.fc.up.pt

Abstract

Public transport systems can be studied as complex networks in which stops are represented as nodes and connections between stops as edges. This work compares the structural properties of the public transport networks of Lisbon, Paris and Luxembourg using data from a public multimodal transport dataset. The networks were modelled as directed graphs and analysed using complementary Network Science measures, including connectivity, degree distribution, clustering coefficient, betweenness centrality and community detection.

The results reveal substantial differences between the three cities. Luxembourg exhibits the highest level of connectivity, with nearly all nodes belonging to a single strongly connected component, a higher average degree and a clustering coefficient approximately one order of magnitude greater than those observed in Lisbon and Paris. In contrast, Lisbon presents the most fragmented network, while Paris occupies an intermediate position. Betweenness centrality analysis indicates that the Luxembourg network relies more strongly on a small number of strategic transport hubs, whereas shortest paths are distributed more evenly across the larger networks. Community detection reveals a strong modular structure in all three cities, although the spatial organization of communities differs considerably.

Overall, the results highlight substantial structural differences between the analysed transport networks and show how highly connected systems may also depend strongly on a small number of strategic transport hubs.

1 Introduction

Public transport systems play a fundamental role in urban mobility by connecting different parts of a city and enabling the movement of people through interconnected transportation networks. Beyond their practical importance, these systems can also be studied as complex networks, where transport stops are represented as nodes and connections between stops are represented as edges. This representation allows the application of Network Science techniques to investigate structural properties such as connectivity, centralization and community organization.

Understanding the structure of transport networks is important because network topology influences accessibility, robustness and the overall organization of mobility systems. Different cities may exhibit substantially different structural characteristics depending on their size, geography and transportation infrastructure.

This work focuses on the comparative analysis of the public transport networks of Lisbon, Paris and Luxembourg. These cities were selected because they represent transport systems of different scales and levels of complexity.

The main research question addressed in this study is:

How do the public transport networks of Lisbon, Paris and Luxembourg differ in terms of their structural properties?

To answer this question, the networks are analysed using a set of complementary Network Science measures, including connectivity, degree distribution, clustering coefficient, betweenness centrality and community structure. The results provide a comparative characterization of the three transport systems and highlight the main structural differences between them.

2 Dataset

This work uses the public transport network dataset presented by Kujala et al. [4]. The dataset contains multimodal public transport networks extracted from GTFS feeds for 25 cities worldwide.

Three cities were selected for this study: Lisbon, Paris and Luxembourg. For each city, two files were used: `network_combined.csv`, containing the combined public transport network of all transportation modes available in that city, and `network_nodes.csv`, containing stop metadata such as names and geographic coordinates.

Lisbon, Paris and Luxembourg were selected because they represent transport networks of substantially different scales and structures. Paris corresponds to a large metropolitan network with multiple transport modes, Lisbon represents a medium-sized multimodal network, and Luxembourg provides an example of a considerably smaller network. This diversity makes it possible to compare how network properties vary across systems with different levels of complexity.

Nodes represent public transport stops and include the stop identifier, geographical coordinates and stop name. Edges represent connections between consecutive stops in transport routes. Each edge also contains additional attributes, including the straight-line distance between stops (`d`, in meters), average travel time (`duration_avg`), number of vehicles traversing the connection (`n_vehicles`) and transportation mode (`route_type`).

The transportation modes differ across the selected cities. Lisbon contains bus, metro, rail and ferry services, Paris contains tram, metro, rail and bus services, while Luxembourg contains rail and bus services only.

A preliminary exploratory analysis was performed to verify data consistency and understand the structure of the dataset. No missing values were found in the analysed files, and all cities shared the same attribute structure.

Table 1 summarizes the main characteristics of the analysed transport networks.

Table 1: Main characteristics of the analysed transport networks.

City	Nodes	Edges	Transport Modes
Lisbon	7073	8982	Metro, Rail, Bus, Ferry
Paris	11950	14781	Tram, Metro, Rail, Bus
Luxembourg	1367	3234	Rail, Bus

3 Methodology

3.1 Network Construction

The transport networks were modelled as directed graphs using the NetworkX library [3]. Each stop was represented as a node and each connection between consecutive stops was represented as a directed edge.

The graph construction was based on the file `network_combined.csv`. Node metadata, including stop names and geographic coordinates, were obtained from `network_nodes.csv` and later incorporated into the graph for visualization and interpretation purposes.

A directed graph (`DiGraph`) representation was adopted in order to preserve the direction of travel between stops. Consequently, a connection from stop A to stop B is treated as distinct from a connection from stop B to stop A.

The original dataset may contain multiple connections between the same pair of stops associated with different transportation modes. Since the `DiGraph` structure does not support parallel edges, duplicate edges were automatically merged into a single directed connection during graph construction.

An exploratory analysis revealed the presence of 142 duplicate edges in Paris and 8 duplicate edges in Luxembourg, while no duplicate edges were found in Lisbon. These duplicated connections correspond to different transport modes serving the same pair of stops rather than data errors. Because they represent less than 1% of the total number of edges in all cases, their impact on the resulting network structure is negligible.

Although the dataset provides attributes related to distance, travel time and service frequency, the analysis was performed on unweighted graphs. This decision was taken to focus on the structural and topological properties of the transport networks and to facilitate direct comparisons between cities.

Pedestrian transfer links available in the dataset were not included in the graph construction. The objective of this study is to compare the structural properties of the public transport networks as represented by direct transport connections. Including pedestrian links would require additional modelling assumptions, such as defining acceptable walking distances between stops, and could affect cities differently depending on their urban density and stop distribution. For this reason, only the transport connections contained in the original combined network files were considered. As a result, network connectivity depends exclusively on direct public transport links.

3.2 Exploratory Data Analysis

Before constructing the graphs, an exploratory analysis was performed to verify data consistency and understand the main characteristics of the selected networks.

The analysis included the inspection of the available attributes, identification of transportation modes, verification of missing values and comparison of network sizes in terms of nodes and edges.

No missing values were found in the analysed files, and all three cities shared the same attribute structure.

3.3 Network Measures

The objective of the analysis is to compare the structural properties of the transport networks of Lisbon, Paris and Luxembourg. To achieve this goal, a set of complementary network measures was selected.

Network connectivity was first evaluated through the analysis of Strongly Connected Components (SCCs). Since the networks were modelled as directed graphs, SCCs provide an appropriate measure of fragmentation by identifying groups of stops that are mutually reachable while respecting edge direction. Particular attention was given to the size of the largest SCC and to the proportion of network nodes it contains.

The distribution of node degrees was then analysed to characterize how transport connections are distributed across stops. For this purpose, the total degree of each node, defined as the sum of its in-degree and out-degree, was considered. Since the objective was to assess overall connectivity patterns rather than directional properties, total degree was used as a simple and comparable measure of local connectivity across the three networks.

Measures based on shortest paths, such as average path length and network diameter, were not included in the comparative analysis. These metrics require paths to exist between all pairs of nodes, a condition that is not satisfied by all analysed networks due to the presence of multiple strongly connected components. Restricting the computation to only the largest strongly connected component would describe only a subset of the network and would reduce the comparability of the results across cities.

Local connectivity patterns were evaluated through the clustering coefficient, which measures the tendency of neighbouring nodes to form interconnected groups.

Node importance was assessed using betweenness centrality. This metric quantifies how frequently a node lies on shortest paths between other nodes and therefore identifies stops that act as important transfer or bridging points within the transport network.

Community structure was analysed using the Louvain community detection algorithm [2], implemented through the Modularity function available in Gephi. The resulting modularity score and the number of detected communities were used to compare how strongly the networks are divided into groups of densely connected stops.

3.4 Network Visualization

Network visualizations were produced using Gephi [1].

The graphs were exported from NetworkX in GEXF format, preserving node attributes such as stop names and geographic coordinates. Geographic layouts were generated using the GeoLayout algorithm, which positions nodes according to their latitude and longitude values.

To facilitate interpretation, nodes were coloured according to their detected community and scaled according to their betweenness centrality values. This representation allows the simultaneous visualization of community structure and the relative importance of transport hubs within each network.

4 Results

4.1 Connectivity Analysis

Table 2: Connectivity indicators for the analysed transport networks.

City	Reverse Edges (%)	SCCs	Largest SCC (%)
Lisbon	3.70	566	36.34
Paris	12.47	2255	73.06
Luxembourg	82.02	7	99.49

Table 2 summarizes the connectivity properties of the three transport networks.

Lisbon and Paris exhibit strongly directed structures, with only 3.7% and 12.5% of edges having an explicit reverse connection, respectively. In contrast, Luxembourg shows a much more symmetric structure, with 82% of edges having a corresponding reverse edge.

The analysis of strongly connected components revealed substantial differences between the networks. Luxembourg contains only seven strongly connected components, with 99.5% of all nodes belonging to the largest component. This indicates that almost the entire network is mutually reachable when edge direction is respected.

Paris presents an intermediate level of fragmentation. Although 2,255 strongly connected components were identified, approximately 73% of all nodes belong to the largest component, indicating the existence of a dominant connected core.

Lisbon is the most fragmented network. Despite containing fewer nodes than Paris, only 36.3% of its stops belong to the largest strongly connected component. The remaining nodes are distributed across hundreds of smaller components, suggesting a substantially lower level of mutual reachability.

Overall, the results indicate a clear connectivity gradient across the analysed cities. Luxembourg is almost entirely strongly connected, Paris exhibits an intermediate structure characterized by a large connected core and several peripheral components, while Lisbon presents the highest degree of fragmentation.

4.2 Degree Distribution

The degree distribution was analysed to characterize how transport connections are distributed across stops in each network. As described in the methodology, the total degree of each node was considered, corresponding to the sum of its in-degree and out-degree.

Table 3: Average degree of the analysed transport networks.

City	Average Degree
Lisbon	2.54
Paris	2.45
Luxembourg	4.72

Table 3 shows the average degree of each network. Figure 1 presents the normalized degree distributions. The distributions were normalized by the total number of nodes in each network, allowing direct comparison between cities of different sizes.

The results reveal clear differences between the three transport systems. Lisbon and Paris exhibit very similar average degrees, indicating comparable levels of overall connectivity. In both cities, the majority of stops have degree two, meaning that most nodes belong to simple linear chains within the transport network.

Luxembourg presents a substantially higher average degree, almost twice that observed in Lisbon and Paris. The degree distribution also shows a larger proportion of nodes with degrees greater than four, indicating the presence of more highly connected stops throughout the network.

The degree distributions of all three cities are strongly right-skewed. Most stops have relatively few direct connections, while a small number of nodes exhibit substantially higher degrees. This pattern is characteristic of many real-world transportation networks, where a limited number of highly connected hubs coexist with a large number of ordinary stops.

These results are consistent with the connectivity analysis presented previously. Luxembourg exhibits both a higher average degree and a substantially larger strongly connected component, indicating a more interconnected network structure. In contrast, the lower average degrees observed in Lisbon and Paris are associated with a predominance of linear transport corridors and a more fragmented topology.

4.3 Clustering Coefficient

The clustering coefficient was used to evaluate the tendency of neighbouring stops to form locally interconnected groups. Table 4 presents the average clustering coefficient for each network.

Table 4: Average clustering coefficient of the analysed transport networks.

City	Clustering Coefficient
Lisbon	0.0075
Paris	0.0069
Luxembourg	0.0726

The results reveal substantial differences between the analysed cities. Luxembourg exhibits a considerably higher clustering coefficient (0.0726), approximately one order of magnitude higher, than Lisbon (0.0075) and Paris (0.0069). This indicates a greater tendency for neighbouring stops to be interconnected, creating locally dense structures within the network.

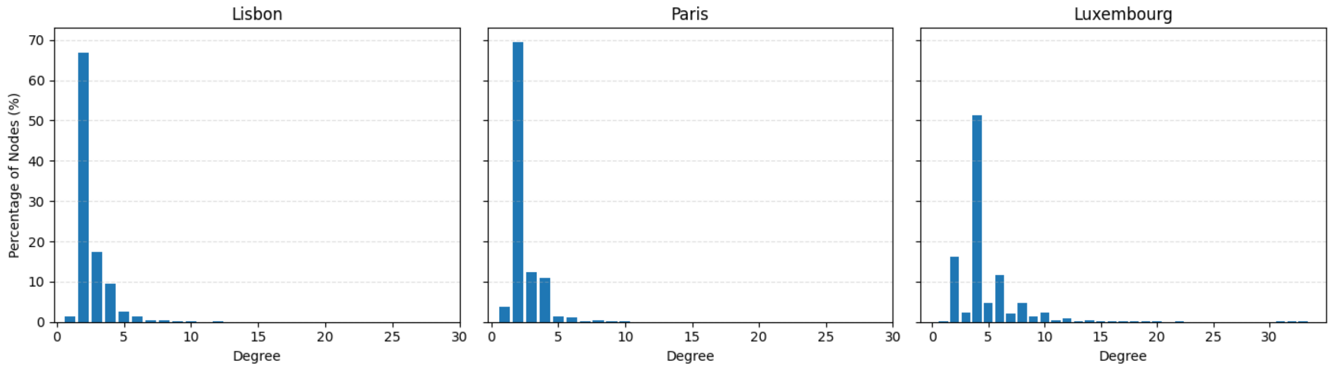

Figure 1: Normalized degree distributions for Lisbon, Paris and Luxembourg. The vertical axis represents the percentage of nodes with a given degree, allowing direct comparison between networks of different sizes.

In contrast, the very low clustering coefficients observed in Lisbon and Paris suggest that most transport connections are organized as linear routes rather than forming highly interconnected local patterns. This observation is consistent with the degree distributions presented previously, where the majority of stops in both cities have degree two.

Together with the connectivity and degree analyses, the clustering results suggest that Luxembourg exhibits a more interconnected network structure, while Lisbon and Paris are characterized by a predominance of linear transport corridors and lower local connectivity.

4.4 Betweenness Centrality

Betweenness centrality was used to identify the most important transport hubs in each network. Since the metric was computed using the normalized formulation provided by NetworkX, the resulting values can be directly compared across cities despite differences in network size.

Tables 5, 6 and 7 present the five stops with the highest betweenness centrality in each network.

Table 5: Top five stops by betweenness centrality in Lisbon.

Stop ID	Stop	Betweenness
6160	Lisboa (Pça Espanha) Terminal P10	0.079
6154	Lisboa (Gare Oriente) Cais 21	0.055
6706	Pragal (Portagem)	0.055
6126	Lisboa (Alcântara) Avª Ceuta Fte Banco Alimentar	0.054
6150	Lisboa (Campolide) Avª C Gulbenkian (Alto)	0.051

Table 6: Top five stops by betweenness centrality in Paris.

Stop ID	Stop	Betweenness
2378	Place de la Chapelle	0.052
2376	Gare du Nord	0.052
2377	Gare du Nord	0.049
4667	Hôtel de Ville	0.048
2428	Place de la Chapelle	0.046

Table 7: Top five stops by betweenness centrality in Luxembourg.

Stop ID	Stop	Betweenness
1393	Luxembourg, Gare Centrale	0.472
268	Bettembourg, Gare	0.227
1349	Kirchberg, John-F.-Kennedy	0.193
1730	Mersch, Gare	0.155
1402	Luxembourg, Gare routière	0.133

The results reveal substantial differences in the concentration of shortest paths across the three transport networks.

Luxembourg exhibits markedly higher betweenness values than Lisbon and Paris. The most central stop, Luxembourg Gare Centrale, reaches a betweenness centrality of 0.472, whereas the highest values observed in Lisbon and Paris are only 0.079 and 0.052, respectively.

This result indicates that a relatively small number of stations concentrate a large fraction of shortest paths in the Luxembourg network. Consequently, these hubs play a particularly important role in connecting different parts of the system.

In contrast, the lower betweenness values observed in Lisbon and Paris suggest that shortest paths are distributed across a larger number of stops, resulting in a less centralized network structure. The connectivity of these networks therefore depends less on individual hubs and is spread more evenly throughout the network.

The identified high-betweenness stops correspond to major transport interchanges and terminal stations, reinforcing their role as key connectors between different regions of the transport network.

4.5 Community Detection and Visualization

Community detection was performed using the Louvain algorithm implemented through the Modularity function available in Gephi. Table 8 summarizes the modularity values and the number of detected communities for each network.

Table 8: Community detection results.

City	Modularity	Communities
Lisbon	0.935	51
Paris	0.938	84
Luxembourg	0.862	27

All three networks exhibit high modularity values, indicating a strong community structure. Lisbon and Paris present very similar modularity scores (0.935 and 0.938, respectively), while Luxembourg exhibits a slightly lower value (0.862). Nevertheless, all values suggest that the transport networks can be partitioned into groups of stops that are more densely connected internally than with the rest of the network. The lower modularity observed in Luxembourg is consistent with its higher level of network integration, since stronger interconnections between regions tend to reduce the separation between communities.

The Louvain algorithm identified 51 communities in Lisbon, 84 in Paris and 27 in Luxembourg. The larger number of communities observed in Paris is consistent with its larger size and greater structural complexity.

The community visualizations reveal important differences in the spatial organization of the three transport systems.

Figure 2 shows the Lisbon network. The detected communities are relatively well separated geographically, with most groups occupying distinct regions of the network. A small number of high-betweenness hubs connect these regions and act as bridges between different communities.

Figure 3 presents the Paris network. Although the modularity value is similar to that of Lisbon, the visualization reveals a considerably more complex structure. Numerous communities overlap within the dense urban core, suggesting a highly interconnected central area, while more geographically distinct communities appear in peripheral areas. This pattern reflects the larger scale and higher complexity of the Paris transport system.

Figure 4 shows the Luxembourg network. Despite containing fewer communities, the network exhibits a highly integrated structure organized around a small number of dominant hubs. This observation is consistent with the connectivity analysis, which showed that nearly all nodes belong to a single strongly connected component, and with the betweenness analysis, which identified a small number of particularly central stations.

Overall, the community detection results indicate that all three transport networks exhibit a strong modular structure. However, the geographic organization of these communities differs substantially. Lisbon presents a more spatially separated organization, Paris exhibits the most complex community arrangement, and Luxembourg appears considerably more integrated around a small number of central transport hubs.

5 Conclusion

This work compared the structural properties of the public transport networks of Lisbon, Paris and Luxembourg using a set of complementary Network Science measures. The analysis revealed substantial differences in network organization, connectivity and community structure across the three cities.

The results showed that Luxembourg exhibits the most connected and integrated network among the analysed cities. Nearly all stops belong to a single strongly connected component, the average degree is considerably higher than in the other networks, and the clustering coefficient indicates a greater tendency for neighbouring stops to form locally interconnected groups. These characteristics suggest a transport network with high overall connectivity and relatively low fragmentation.

In contrast, Lisbon presents the most fragmented structure. Despite containing fewer stops than Paris, only 36.34% of its nodes belong to the largest strongly connected component. Paris occupies an intermediate position, exhibiting higher connectivity than Lisbon but remaining substantially more fragmented than Luxembourg. These results suggest that network size alone does not fully explain connectivity patterns. Although Luxembourg is the smallest network analysed, it exhibits the highest levels of structural integration among the three cities.

The betweenness centrality analysis revealed an additional structural characteristic of the Luxembourg network. While the network is highly connected, a relatively small number of stations concentrate a large proportion of the shortest paths. In particular, Luxembourg Gare Centrale exhibits a substantially higher betweenness centrality than any stop identified in Lisbon or Paris. This suggests that the Luxembourg network depends more strongly on a small number of strategic transport hubs. In contrast, shortest paths in Lisbon and Paris are distributed across a larger number of stops, resulting in a less centralized structure.

Community detection further highlighted differences in network organization. All three networks exhibited high modularity values, indicating a strong community structure. However, the geographic distribution of these communities differed considerably. Lisbon presented a more spatially separated organization, Paris exhibited the most complex arrangement of communities, and Luxembourg appeared more integrated around a small number of central hubs.

The results should be interpreted as properties of the transport network topology rather than the complete passenger mobility system. The analysed graphs include only direct public transport connections between stops. In practice, passengers may walk between nearby stops, creating additional connections that are not represented in the networks considered in this study. Such pedestrian transfers would likely increase connectivity and reduce the observed level of fragmentation, particularly in the larger urban networks.

Future work could incorporate pedestrian transfer links available in the original dataset in order to investigate how walkable connections affect network connectivity, accessibility and community

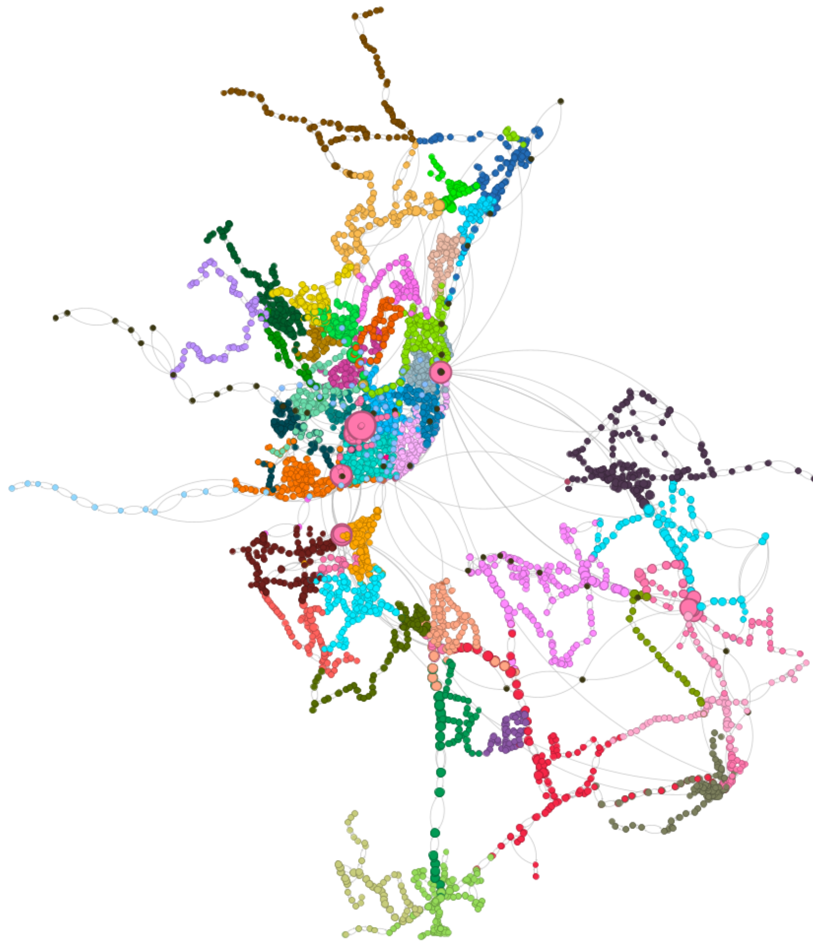

Figure 2: Community structure of the Lisbon transport network. Node colours represent Louvain communities and node size is proportional to normalized betweenness centrality.

structure. Additional analyses could also explore weighted versions of the networks using travel times, distances or service frequencies to better capture operational characteristics of the transport systems.

References

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Figure 3: Community structure of the Paris transport network. Node colours represent Louvain communities and node size is proportional to normalized betweenness centrality.

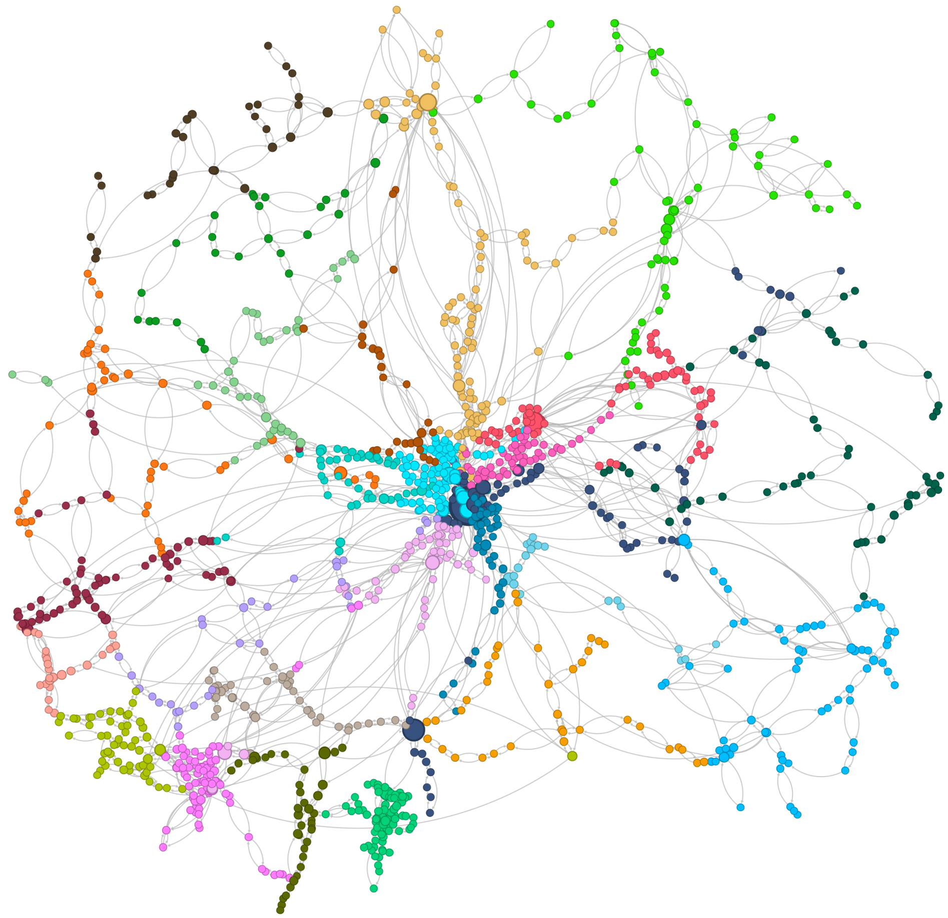

Figure 4: Community structure of the Luxembourg transport network. Node colours represent Louvain communities and node size is proportional to normalized betweenness centrality.